SOFTWARE SYSTEMS

1. a) Invent an example of a composite attribute you might find in a real-world data model, giving both the composite name and the sub-parts. You cannot use addresses or names as your example. [5]

Answer:
We could use "dimensions", which splits into "width" and "height".

b) Invent an example of a foreign-key relationship in a real-world data model. You cannot use relationships involving people as your example. [5]

Answer:
We could use LocatedIn to describe the relationship between a circuit board (primary) and the component located on the circuit board (foreign key from component to circuit board).

c) How many rows and columns will the query `SELECT * FROM orders WHERE ProductId = 'xt23341';` produce? [5]

Answer:
It will produce the same number of columns as the orders table has. The number of rows will be determined by how many rows have the matching ProductId.

d) What SQL operator is closest to the $\rtimes\!\bowtie$ operator, and what would $A \rtimes\!\bowtie B$ return if B contains no rows? [5]

Answer:
The closest is LEFT JOIN, or LEFT OUTER JOIN. If B is empty then the result of $A \rtimes\!\bowtie B$ contains all the rows of A, padded with empty columns for the extra attributes than are in B but not in A.

e) Translate these two C structures into equivalent relations. [5]

```
struct Company{
  string name;
};
struct Component{

  string key;
  // 0 or more companies may make a component
  vector<Company *> manufacturers;
}
```

Answer:

- *Company(name)*
- *Component(key)*
- *Manufacturers(name,key)*

Answers

The main insight is that the manufacturers relationship cannot be expresses as just a single primary-foreign key relationship between Company and Component.

f) State whether the following statement is true or false, and briefly explain why: "HTTP can only work over Ethernet networks." [5]

Answer:
This is false. An example of a non-EEthernet network would be browsing the internet while using bluetooth tethering, which is not ethernet based. Some aspects (e.g. connection upgrades or HTTP 3) might assume IP, but clearly this can still run over non-Ethernet channels.

g) If a computer does not have access to a DNS server, can it still send and receive IP packets? Explain your answer. [5]

Answer:
Yes, it can still send and receive IP packets. DNS is not require in order to route IP packets, and is only needed to map domain names to IP addresses.

h) Give one theoretical advantage of a packet-switched network over a circuit-switched network, and give a concrete example of where that advantage matters in practice. [5]

Answer:
In a packet-switched network it is possible for data-flows to follow a different route if a channel is disrupted. An example of this would be traffic moving between IP addresses in Europe and the US – if transatlantic cables go down, then data can be re-routed via other routes, for example via Asia, but (in principle) the two endpoints do not need to do anything/

i) What is special about IP port numbers less than 1024, and give an example of a piece of software which binds to a port below 1024. [5]

Answer:
These are "well-known" port numbers where connections to specific services are made. An example of a piece of software binding below 1024 is an SSH server.

j) What is the purpose of the checksum field in an IP packet? [5]

Answer:
The checksum is used to detect corruption of the packet as it is moved from the source to the destination. If the packet is corrupted it can be dropped before reaching the destination.

Answers

2. You have been asked to design the network and data architecture for a system containing very large numbers of low-power sensor nodes. Each sensor node is able to determine both its position and the current time using GPS, and can measure the local temperature, pressure, and humidity through on-board sensors. A major constraint on the nodes is battery life – the nodes have a small solar panel and battery, and due to power constraints must spend a lot of time sleeping.

The sensor nodes contain a wireless network interface which allow them to send and receive packets over long distances, though this can only happen infrequently due to the power constraints. All nodes are within communication range of at least one server, each of which is a standard server with a mains power supply. All servers know their own GPS location, and can communicate with each other using a wired network.

a) Each server has no initial knowledge about the number and positions of nodes and other servers, and must build a model of them as it receives sensor readings from them.

i) Identify three entity sets within this system. [3]

Answer:
Examples could be "servers", "nodes", and "readings".

ii) Identify a natural primary key within the system, stating any assumptions you make. [3]

Answer:
Each node probably has a serial number or other embedded unique id such as a MAC address. This assumes that the nodes are not pure clones of each other.

iii) Draw an ER diagram of the system containing your three entity sets, and at least one relationship set involving each entity set. [6]

Answer:
A simple example would be:

```
                N                M          1            N
[Servers]---<ReachableFrom>---[Nodes]---<TakenBy>===[Readings]
```

iv) Describe a many-to-many relationship set in the system. This may or may not be a relationship set shown in your ER diagram. [3]

Answer:
Each node may be reachable from multiple servers, and each server may be able to reach many nodes.

b) The servers are connected via a wired network, but do not initially know how many other servers there are or their addresses.

i) Assume the servers are connected together by a single Ethernet switch, but are not connected to the global internet. How could the servers find each other? [4]

Answer:

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If they are on the same Ethernet switch, then they can be on the same LAN. Broadcasts at the LAN level will reach all servers, so the servers can find each other using broadcasts on the LAN.

ii) Now assume the servers are on different LANs, but are all connected to the global internet. How could the servers find each other? [4]

Answer:
It is not possible to use broadcasts, but it becomes possible to create a well-known registration point. For example, the servers could all contain the DNS name of a service, and when they start up they register with the service, and then receive back a list of all other registered servers.

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c) By default the nodes try to report their latest sensor readings every hour on the hour, using the GPS clock to work out when to send, and then sending a single packet using a connection-less protocol. This works fairly well for small test deployments using $n = 100$ nodes, but when deployed with $n = 1000$ nodes many updates are lost, and for $n = 10000$ nodes very few update packets get through due to packet loss.

i) Explain the problem that is occurring. [3]

Answer:
Each node is trying to broad cast over a shared medium, and due to the lack of access control they are trying to send at the same time - probably this timing is quite precise due to the GPS clock. Each pairwise conflict results in the loss of the packet, and as the number of nodes increases the probability of a conflict also increases.

ii) Develop a simple mathematical model to estimate the rate of packet loss as a function of the number of nodes. Describe any extra parameters needed and any assumptions made. This model does not need to be exact, but should quantitatively explain why the problem becomes substantially worse as n increases. [5]

Answer:
Assume that if two nodes try to send there is a probability p that they manage to overlap by sending at exactly the same time. If there are three nodes, then there are three pair-wise conflicts, and one three-way conflict. At its most simplistic, if we consider n nodes, then the probability that node 1 doesn't conflict with any other packets is $(1-p)^{n-1}$*, then for node 2 it is* $(1-p)^{n-2}$ *and so on. Even just taking the first term of* $(1-p)^{n-1}$ *we can see that even if p is small, the probability of the packet getting through degrades quickly with higher n. The probability of* all *packets getting through is* $(1-p)^{n-1}(1-p)^{n-2}(1-p)^{n-3}\ldots$ *which very rapidly decreases.*

No specific mathematical or probabilistic model is needed here, as long as it identifies clearly in some way why the problem gets much worse for large n.

iii) If the goal is simply to receive one sensor reading per hour *on average*, and the sensor node energy budget only allows the sending of one packet in each hour, propose a change to the sending strategy that increases the chances of each sensors hourly update packet being received. [4]

Answer:
A simple approach is to randomly offset each nodes sending time within the hour. This could be a fixed offset per node, or something chosen on an hour by hour basis. To get the best possible utilisation this would need to be co-ordinated, but to get round the problem of all nodes trying to send small packets at exactly the same time this should work well.

iv) The goal is now to try to capture all sensor readings as close to the hour as possible, but the energy budget does allow more than one packet to

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be sent per hour. Propose a new sending strategy for this use-case. [4]

Answer:
Each node could try to send their packet on the interface as normal, then use sensing to determined if the packet was corrupted by another sensor. If a collision happens, then random exponential backofff can be used (as in ethernet) to adapt to the number of contending nodes.

d) Assume that the data received from the sensors has been post-processed and aggregated into distinct regions, and exported to two tables with the following structure:

```
CREATE TABLE regions (
    id STRING PRIMARY KEY,
    x FLOAT,
    y FLOAT
);
CREATE TABLE daily_samples (
    region_id STRING FOREIGN KEY REFERENCES regions(id)
    date DATE,
    temperature FLOAT,
    pressure FLOAT,
    humidity FLOAT
);
```

Note: these tables are not directly related to the data model described earlier, and would not be expected to appear in the ER model.

i) Give an SQL query that counts the number of regions. [3]

Answer:
```
SELECT COUNT (*) FROM regions;
```

ii) Give an SQL query that returns the average temperature on a given date across all regions. [4]

Answer:
```
SELECT avg (temperature) FROM daily_samples WHERE data='some-date';
```

iii) If there are N regions and M readings and every reading refers to a valid region, what is the maximum number of rows if we join the two tables on the region id? [4]

Answer:
Each reading is associated with 1 region, so the number of readings cant change. SO the total number of readings will remain M.

Answers